

PRANJAL SWARUP BHATNAGAR

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EXPERIENCE

Associate, Norman Lab

Next Tech Lab (QS Reimagine Education Award winner)

Apr 2026 - Present
SRMIST, Kattankulathur

- Selected by interview into Norman Lab at Next Tech Lab, the first student-led organisation from India to win the QS Reimagine Education Award.
- Built StupidHacks, the portal for the lab's internal hackathon, with a small team. Owned the data layer and its rule tests, the design system on the lab's brand, and the move to Supabase with one-time-code roster sign-in. Live at stupidhack.nexttechlab.in.

PROJECTS

ExamDB | Python, FastAPI, PostgreSQL, Next.js, Docker

- Built and shipped a searchable database of **194** Indian competitive exams, consolidating dates scattered across dozens of government portals into one source that cites the official page behind every date.
- Designed a scrape/normalise/dedup/reduce pipeline that preserves conflicting sources and flags the disagreement instead of silently picking a winner, so wrong data stays traceable to the scraper that produced it.
- Caught a scraper that passed its own tests while emitting **46** rows of a historical archive stamped with the current year, spotted from an impossible 29 FEB entry. Added source-diff validation so a row count is never mistaken for correctness.
- Built OAuth sign-in storing only SHA-256 digests of opaque session tokens, plus daily scrapers on GitHub Actions and a Next.js frontend. Solo project, live at examdb.org.

TYPE | Rust, ratatui, LSP, DAP, tree-sitter

- Building a full IDE that runs in the terminal, non-modal with full mouse parity, aimed at the gap between bare terminal editors and heavyweight GUI ones. Published on crates.io as typ-editor, currently **v0.2.10**.
- Wrote LSP, DAP and tree-sitter clients so new language support arrives by protocol instead of per-language code. Holds a **100ms** cold-start and **16ms** p99 keystroke-to-glyph budget.
- Added an opt-in vim layer that maps modes and operators onto the same named actions the non-modal path already calls, so it owns no editing primitives and the core never depends on it.
- Made themes a TOML artifact rather than a compile-time constant, detecting terminal colour depth at startup and degrading the palette to 256 colours where required.

OutN | Python, discord.py, OpenCV

- Self-hostable FOSS Pokemon recognition and assistant bot for Discord, identifying spawns from images and solving hints automatically; config-driven so users deploy it without writing code.
- Maintained across **2+ years** and **9 major versions** through repeated breaking changes to the upstream Discord and Poketwo APIs, keeping existing self-hosted deployments working without manual migration.

EDUCATION

SRM Institute of Science and Technology

2025 - 2030

M.Tech (Integrated) Computer Science, Cybersecurity & Digital Forensics

Delhi Public School

2011 - 2024

Class XII

SKILLS

Languages	Python, Rust, TypeScript, JavaScript, Go, SQL
Backend	FastAPI, SQLAlchemy, PostgreSQL, Hono, Express
Frontend	Next.js, React, Svelte, Astro, TailwindCSS
Infrastructure	Docker, GitHub Actions, Vercel, Supabase, Redis, Git